Toward Seamless 24-Hour Monitoring of Surface Ozone Across China via Satellite Observations and Sample-Weighted Machine Learning

Yuenong Su, Tongwen Li, *Member, IEEE*, and Jingan Wu

***Abstract*—Ozone (O_3) pollution has been worsening in China in recent years. Hourly O_3 estimation from geostationary satellites plays a crucial role in pollution monitoring and early warning. However, two major challenges remain. First, O_3 exhibits diurnal variations, yet existing studies typically focus only on daytime estimation, resulting in a lack of nighttime O_3 data, which limits the understanding of day-night variations. Second, current research often suffers from the underestimation of high O_3 concentrations, but few studies have adequately addressed this issue. Therefore, we aim to generate seamless 24-h O_3 concentrations across China while effectively mitigating the underestimation of high O_3 concentrations. Specifically, separate daytime and nighttime inversion models are developed using shortwave radiation (SWR) and surface temperature data as the primary inputs, respectively, and the outputs from these models are then fused to produce spatially and temporally continuous 24-h O_3 data. Meanwhile, a sample-weighted machine-learning approach is proposed to alleviate the underestimation of high O_3 values by assigning greater weights to high-value samples. Validation results show that the daytime (nighttime) model achieves an R^2 of 0.89 (0.83) and an RMSE of 13.76 $\mu g/m^3$ (16.76 $\mu g/m^3$) for the site-based cross-validation (CV). The sample-weighted method improves the R^2 value of the daytime model (nighttime model) for high-value samples by 6.4% (13.2%), and reduces the RMSE by 13.9% (24.4%). Our study reveals a 16.4% bias in the daily average O_3 calculation when nighttime O_3 is not taken into account. In this study, seamless 24-h O_3 data across China are estimated based on satellite observations and sample-weighted machine learning, which has a broad application prospect in O_3 pollution monitoring.**

***Index Terms*—24-hour estimation, Ozone (O_3), remote sensing, sample-weighted machine learning.**

I. Introduction

GROUND-LEVEL ozone (O_3) pollution poses significant risks, which not only irritates the eyes and respiratory tract and contributes to cardiovascular, cerebrovascular, and respiratory diseases [1], [2], [3], but also adversely affects key crops such as rice and wheat [4], [5]. In addition, O_3 affects local precipitation patterns and atmospheric circulation [6], [7], and also acts as a greenhouse gas, intensifying the global greenhouse effect and impacting the global climate [8], [9]. Numerous studies have shown that O_3 pollution in China has been on the rise in recent years. In 2019, the average MDA8 (Maximum Daily 8-h Average) O_3 concentration in the North China Plain reached 83 ppb, with the maximum value of 129 ppb [10]. Furthermore, the average 90th percentile MDA8 O_3 concentrations in the provincial capitals of mainland China rise by 23% [11]. These trends underscore the urgent need for comprehensive monitoring of surface O_3 pollution across China [12].

Satellite remote sensing enables large-scale observation of surface O_3 concentrations. Two primary types of satellites are typically used for O_3 estimation: polar-orbiting satellites and geostationary satellites. Polar-orbiting satellites have longer revisit cycles, which generally observe O_3 concentrations at a daily scale. Early studies that employ satellite remote sensing to estimate surface O_3 concentrations primarily rely on polar-orbiting satellites and focus on daily-scale variations in O_3 concentrations [13], [14]. For example, Li et al. [15] estimate MDA8 O_3 concentrations in Tibet, China, from 2005 to 2018, using column O_3 data from the OMI sensor. Zhu et al. [16] developed a coupled surface O_3 estimation framework (LESO) to estimate daily O_3 concentrations from TROPOMI data. Furthermore, Zhu et al. [17] retrieved daily surface O_3 concentrations, along with five other major air pollutants, across mainland China from 2019 to 2020 using TROPOMI satellite observations. However, daily-scale O_3 data fail to capture diurnal variations in O_3 concentrations, limiting their effectiveness in capturing pollution events.

In contrast, geostationary satellites provide hourly or even higher temporal resolution observations, making them effective tools for monitoring the diurnal variation of O_3. In recent years, researchers have increasingly used geostationary satellite observations to estimate surface O_3 concentrations, aiming to improve the temporal resolution of inversion results from a daily scale to an hourly scale. For instance, Wang et al. [18] employ multiple infrared and thermal-infrared bands from the Himawari-8 satellite to estimate hourly surface O_3 concentrations over China (09:00 to 18:00 local time) at a 2-km resolution from 2017 to 2019. Similarly, Chen et al. [19] use O_3-sensitive spectral bands from Himawari-8 along with the random forest model to invert hourly (10:00 to 15:00 local time) surface O_3 concentrations at a 5-km resolution from September 2015 to August 2021. Additionally, Zhang et al. [20] utilize the shortwave radiation (SWR)

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TABLE I
SUMMARY OF DATA USED IN THE STUDY

Data	Variable	Name	Unit	Resolution		Source
				Temporal	Spatial	
In-situ Data	O_3	Ozone	$\mu g/m^3$	hourly	*In situ*	CNEMC
	SWR	Shortwave radiation	W/m^2			
Satellite Data	UVA	Ultraviolet-A radiation	W/m^2	hourly	$0.05° \times 0.05°$	Himawari-8
	UVB	Ultraviolet-B radiation	W/m^2			
LST Data	LST	Land surface temperature	K	hourly	$0.05° \times 0.05°$	GHA-LST
Auxiliary Data	U-Wind	10m u-component of wind	m/s	hourly	$0.25° \times 0.25°$	ERA5
	V-Wind	10m v-component of wind	m/s			
	T2M	2m temperature	K			
	BLH	Boundary layer height	m			
	E	Evaporation	m of water equivalent			
	STRD	Surface thermal radiation downwards	J/m^2			
	R	Relative humidity	%			
	HCHO	Formaldehyde	kg kg^{-1}	3-hourly	$0.75° \times 0.75°$	EAC4
	NO_2	Nitrogen Dioxide	kg kg^{-1}			

product from Himawari-8 as the primary input for the bagged-tree model to estimate hourly (09:00 to 16:00 local time) surface O_3 concentrations at 5-km resolution from March 2018 to February 2019. In these studies, machine-learning methods have become increasingly prominent due to their ease of implementation, high computational efficiency, and strong predictive performance. Commonly used approaches include Random Forest [21], XGBoost (eXtreme Gradient Boosting) [22], LightGBM (Light Gradient Boosting Machine) [23], and deep neural networks [24].

However, two major limitations remain in the current research of satellite-based hourly O_3 estimation. First, O_3 pollution exhibits significant diurnal variation, yet many studies only cover daytime hours (approximately 09:00 to 16:00 local time), leaving substantial gaps in nighttime O_3 data. Nighttime O_3 pollution can harm human health [25], affect the growth of plants and crops [26], and accumulate to contribute to elevated O_3 concentrations during the following day [27]. Therefore, the absence of nighttime O_3 pollution data undermines the assessment of nighttime O_3 hazards and severely hinders a comprehensive understanding of day-night O_3 patterns. Meanwhile, many of these studies rely on data from the Himawari-8 satellite, which cannot observe the area west of 80°E in China [28], [29]. As a result, the O_3 estimates from these studies have geographic gaps, posing challenges for users who require complete spatial coverage. Second, there is a frequent issue of high-value underestimation when using machine-learning methods for O_3 estimation; that is, the machine-learning models tend to underestimate high-value O_3 concentrations [30], [31]. From the perspective of O_3 pollution monitoring, high concentrations are of particular concern. Nonetheless, relatively few studies have addressed the underestimation of high values in machine learning-based O_3 inversion, and how to improve model performance for high values remains an open and critical research question.

To address the above problems, we construct a model with Himawari-8 SWR data as the main input data to invert the surface O_3 concentrations data in China during daytime, and establishes another model with surface temperature data as the main input data to make up for the lack of the inversion results in the nighttime and the area west of 80°E in China, so as to obtain seamless O_3 data for 24-h through data fusion. Meanwhile, a sample-weighted machine-learning model is established to assign greater weights to high-value samples in the model to amplify the importance of high-value samples, thereby mitigating the high-value underestimation phenomenon. The proposed model is tested by sample-based and site-based ten-fold cross-validation (CV). Finally, this study produces the 5-km resolution seamless 24-h surface O_3 data for China in 2021, and the distribution and changes of surface O_3 concentrations in China are investigated.

II. STUDY DATA

This study uses SWR product data from Himawari-8, surface temperature data obtained from GHA-LST, meteorological data provided by ERA5 reanalysis data, and O_3 precursor data from EAC4 reanalysis data as input data. Meanwhile, site monitoring surface O_3 concentrations data provided by the China National Environmental Monitoring Center (CNEMC) are used as the true O_3 concentrations data to train the model. The data used covers the period from January 1, 2021, to December 31, 2021. The summary of the data is shown in Table I.

A. O_3 Site Observation

Nearly 1600 monitoring sites set up by CNEMC in mainland China can provide hourly air pollutant data, including surface O_3 concentrations. The distribution of the sites is shown in Fig. 1, which is denser in the eastern, southern, and central regions of China and sparser in the western and northern regions of China. In this study, outliers

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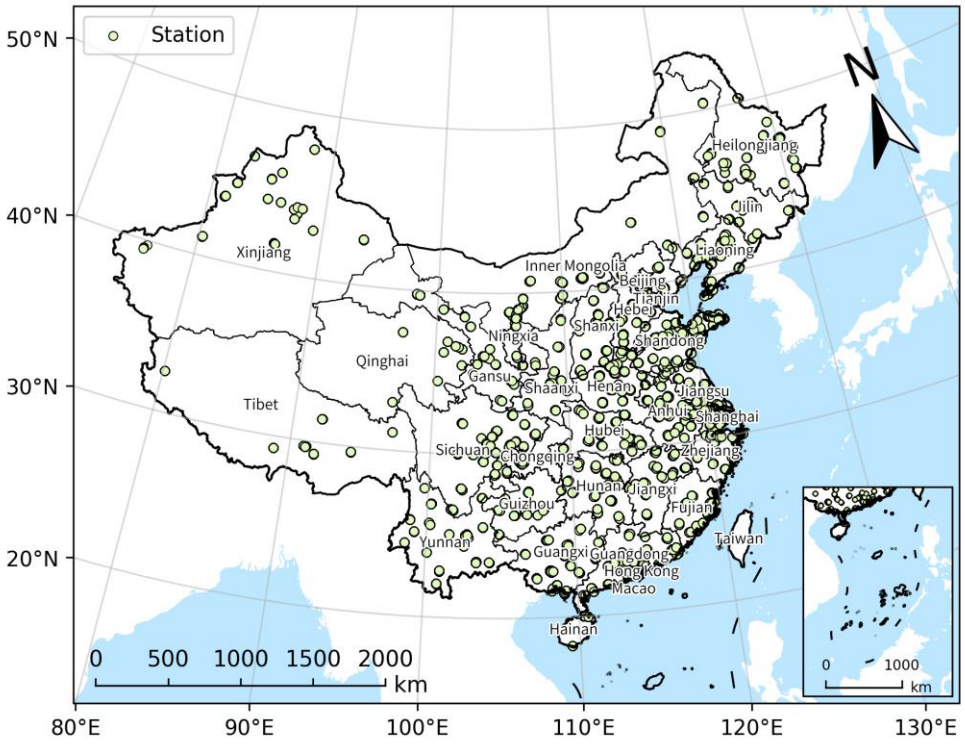


Fig. 1. Spatial distribution of ground O_3 monitoring sites in China.

(e.g., values > 1000 $\mu g/m^3$ or equal to 0) are excluded from the analysis. Statistical results of the site observations show that the average daytime (defined as 09:00–16:00 local time for consistency with the subsequent modeling) O_3 concentration across China is 75.3 $\mu g/m^3$, and the average nighttime O_3 concentration is 51.9 $\mu g/m^3$, exhibiting a clear diurnal difference.

B. Himawari-8 SWR Data

Photochemical reaction is an important source of surface O_3 production, and the amount of O_3 produced by this reaction has an important relationship with the intensity of solar SWR [32]. This study uses the SWR data produced by the satellite Himawari-8, launched by the Japan Meteorological Agency (JMA) [33], as a main data source, which can provide ultraviolet-A radiation (UVA), ultraviolet-B radiation (UVB) data, and SWR data [34]. Since the SWR data is almost seamless, the inversion of surface O_3 using this data will not have any missing results due to cloud cover [20]. The SWR product has a temporal resolution of 1 h and a spatial resolution of 0.05° × 0.05°. However, because of the observation range of the satellite Himawari-8, it cannot observe the area west of 80°E of China [35].

C. Surface Temperature Data

Temperature is one of the important factors affecting the reaction rate, and higher temperatures will accelerate the rate of the photochemical reaction and increase the O_3 concentrations [36]. Temperature also influences boundary layer structure and mixing conditions (especially at night), thereby affecting the transport and dispersion of O_3. It has been demonstrated that exploiting the relationship between surface temperature and O_3 is an effective approach for retrieving O_3 concentrations during both daytime and nighttime [37], [38]. The surface temperature data used in the study are the GHA-LST (Global Hourly All-sky-LST) dataset derived from satellite remote sensing by Jia et al. [39]. This dataset can provide the global 24-h 0.05° × 0.05° resolution surface temperature data from 2011 to 2023.

D. ERA5 Reanalysis Data

Surface O_3 production, transportation, and deposition are related to meteorological conditions such as temperature, humidity, wind direction, and speed [40], and the ERA5 reanalysis data released by the European Center for Medium-Range Weather Forecast (ECMWF) can provide global 24-h meteorological data in a long time series [41]. The datasets "ERA5 hourly data on single levels from 1940 to present" [42] and "ERA5 hourly data on pressure levels from 1940 to present" [43] are used as inputs to the model, including the u-component (U10) and v-component (V10) of winds, as well as temperature (T2M), boundary layer height (BLH), evapotranspiration (E), relative humidity (R), and downward surface thermal radiation (STRD) parameters, with a spatial resolution of this data is 0.25° × 0.25°.

E. EAC4 Reanalysis Data

Formaldehyde (HCHO) and nitrogen dioxide (NO_2) are precursors in the photochemical reaction of O_3, and using them as model inputs allows us to more fully learn the relationship between precursors and O_3 in the photochemical reaction [44],

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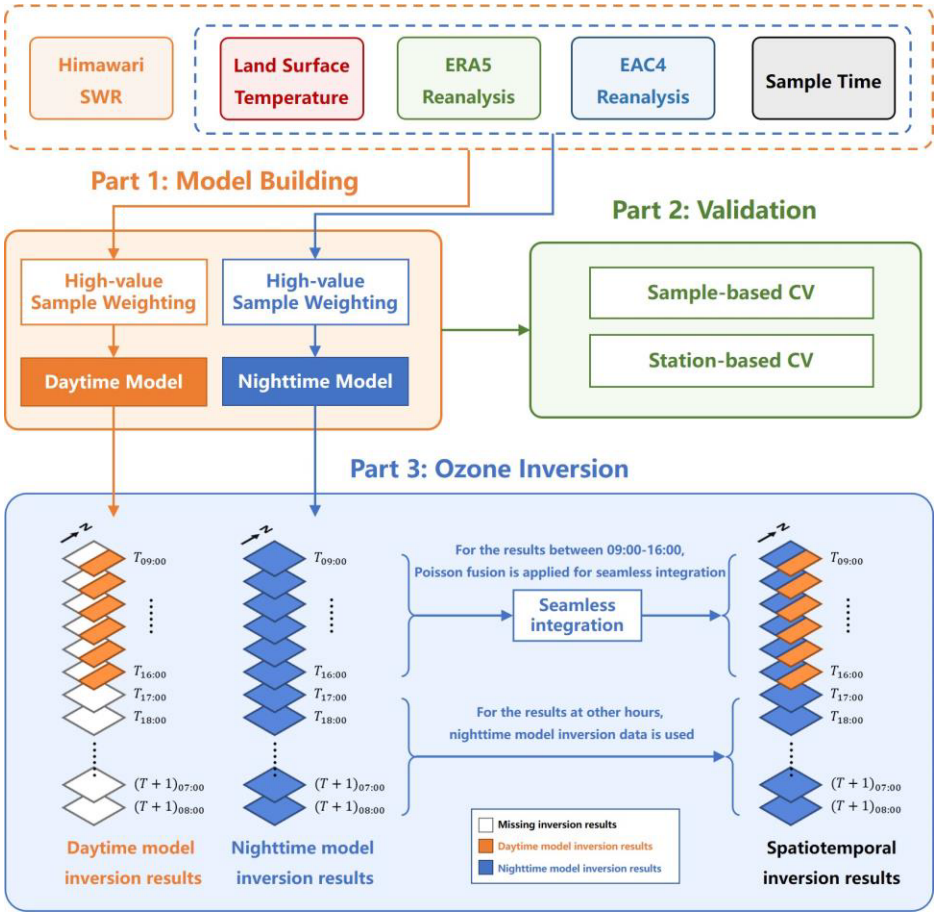


Fig. 2. Flowchart of the seamless 24-h O_3 estimation framework.

[45]. HCHO and NO_2 data from the Copernicus Atmosphere Monitoring Service (CAMS) EAC4 reanalysis [46] are used. The spatial resolution of the reanalysis data is $0.75° \times 0.75°$.

F. Data Pre-Processing

To ensure consistent spatial resolution across all input datasets, the ERA5 and EAC4 data are resampled to a spatial resolution of $0.05° \times 0.05°$ using bilinear interpolation. In addition, the EAC4 data, originally provided at a 3-h temporal resolution, are resampled to an hourly resolution using nearest-neighbor interpolation. Furthermore, satellite data and auxiliary datasets are matched to ground-based observations based on their spatial locations and time to construct the training and evaluation sample sets for the model.

III. Seamless 24-h O_3 Estimation With Sample-Weighted Machine Learning

The general framework of our study is shown in Fig. 2. First, we construct a daytime model with Himawari-8 SWR data as the main data, and a nighttime model with surface temperature data as the main data, and deal with the underestimation of high values when constructing the models. Next, the constructed daytime and nighttime models are evaluated based on the sample-based CV and site-based CV. Subsequently, we use the daytime model to invert the surface O_3 concentrations in the area east of 80°E in China during the daytime hours (09:00-16:00 local time), and use the nighttime model to invert the surface O_3 concentrations in the area west of 80°E in China during the daytime hours, as well as the surface O_3 concentrations in the whole China during the nighttime hours (00:00-08:00 and 17:00-23:00 local time). Finally, the inversion results of the daytime model and the nighttime model are integrated, in which the inversion results of the two models in the daytime period are seamlessly fused, while the inversion results of the nighttime model are directly used in the nighttime period. As a result, the seamless 24-h surface O_3 concentrations for the whole of China are obtained.

A. Daytime and Nighttime Inversion Models

Because of the high temporal resolution (hourly) and large geographical range (China), the dataset used for training is large (about 10 million samples), and we may face the problem of insufficient memory and slow training speed during training. LightGBM is an efficient gradient boosting framework that uses a histogram algorithm, one-sided gradient sampling, mutually exclusive feature bundling, and a leaf-wise algorithm with depth limitation to optimize the training speed and memory consumption of the model [23]. LightGBM has been extensively employed in the inversion of surface O_3 concentrations and has exhibited strong performance across various studies [47], [48]. Therefore, we use LightGBM as

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the machine-learning framework for daytime and nighttime modeling.

Solar radiation plays an important role in the photochemical reactions that generate O_3 and is necessary for photochemical inversion [49], [50]. Solar SWR has become one of the key parameters in the estimation of surface O_3 concentrations [20]. The daytime model of our study is based on the SWR data of Himawari-8 during the period of 09:00-16:00 local time as the main input data, and the surface temperature data, the reanalysis data of ERA5 and EAC4 as well as the temporal information (month, day, and hour) of each sample as the auxiliary data, and the site monitoring O_3 concentrations data are used as the true value to train LightGBM [as shown in the following equation]:

$$O_3\text{daytime} = f_{\text{daytime}}\left(\text{SWR, LST, ERA5,EAC4,month, day, hour}\right) \quad (1)$$

where O_3daytime represents the surface O_3 inversion results from 09:00 to 16:00 local time, f_{daytime} represents the daytime model, SWR represents the Himawari-8 SWR data, LST represents the surface temperature data, ERA5 represents the ERA5 reanalysis data, EAC4 represents the EAC4 reanalysis data, and month, day, and hour represent the sample monthly, daily, and hourly information, respectively.

However, the inversion results of surface O_3 concentrations at night are missing due to the fact that the SWR data of Himawari-8 are not available. In addition, the inversion results of surface O_3 concentrations during daytime do not fully cover China because the Himawari-8 satellite is unable to observe the area west of 80°E in China. To solve this problem, a nighttime model without Himawari-8 SWR data is constructed to fill the gap in the daytime model inversion results. It has been shown that the estimation of surface O_3 concentrations using surface temperature is one of the effective ways for both daytime and nighttime O_3 inversion [37], [38]. Therefore, our study constructs the nighttime model $f_{\text{nighttime}}$ [shown in the following equation] by taking the surface temperature data of the whole day and 24-h as the main input data, and the reanalysis data of ERA5 and EAC4, as well as the temporal information (month, day, and hour) of each sample as the auxiliary input data

$$O_3\text{nighttime} = f_{\text{nighttime}}\left(\text{LST, ERA5,EAC4,month, day, hour}\right) \quad (2)$$

where O_3nighttime represents the inversion results of the nighttime model $f_{\text{nighttime}}$. Finally, our study uses the constructed daytime model to invert the surface O_3 concentrations in the area east of 80°E in China from 09:00 to 16:00 local time, and the nighttime model to invert the surface O_3 concentrations in the whole of China from 00:00 to 08:00 and from 17:00 to 23:00 local time, so as to make up for the lack of the inversion results of the daytime model in nighttime hours. Meanwhile, the nighttime model can invert the surface O_3 concentrations in the area west of 80°E in China from 09:00 to 16:00 to make up for the lack of the daytime model during the spatial period because the input data used in the nighttime model covers 24 h.

We construct the daytime and nighttime models separately to obtain seamless 24-h O_3 data in China. The parameters used in the models are summarized in Supplementary Table S1. However, the daytime model and the nighttime model are two models trained with different input data, and their inversion results may show obvious splicing traces if they are directly spliced. In this regard, the solution adopted in the study is to use Poisson fusion [51] to process the splices during the daytime hours to ensure the spatial consistency of the O_3 concentrations inversion results are smoothed (Supplementary Fig. S1).

B. Mitigating High-Value Underestimation With Sample Weighting

When using machine-learning models to invert the concentrations of atmospheric pollutants such as O_3, it is common to encounter the issue of underestimating high O_3 concentrations, which is possibly caused by the imbalance in the number of high-value samples and low-concentration samples in the dataset. The machine-learning model is trained to minimize the overall residuals, and mainly fits a larger number of low-concentration samples. To solve this problem, we propose a sample-weighted approach by assigning higher weights to high-value samples to enhance their influence on the model. In our study, samples with concentrations exceeding 100 $\mu g/m^3$, which corresponds to the MDA8 O_3 Class I limit specified in the Ambient Air Quality Standard (GB 3095–2012), are given a weight of 5. The value of the weight is derived from the sensitivity analysis presented in Supplementary Table S2. All other samples are assigned a weight of 1. This weighting strategy is implemented by passing the weights to the "sample_weight" parameter in the fit function of the LGBMRegressor.

IV. Results and Discussion

A. Validation Performance

1) Model CV Performance: Fig. 3 shows the overall performance in the sample-based and site-based CV of the daytime and nighttime models. The daytime model performs well in both sample-based and site-based CV, with an R^2 (RMSE) of 0.92 (11.92 $\mu g/m^3$) in the sample-based CV and an R^2 (RMSE) of 0.89 (13.76 $\mu g/m^3$) in the site-based CV. The nighttime model reaches R^2 (RMSE) of 0.87 (14.58 $\mu g/m^3$) in the sample-based CV and R^2 (RMSE) of 0.83 (16.76 $\mu g/m^3$) in the site-based CV. The higher performance in the sample-based CV compared to the site-based CV may be due to the more stringent validation method used in the site-based validation, in which spatially independent sites are selected for testing [52]. Compared with the nighttime model, the daytime model shows better performance. As shown in Supplementary Fig. S2, the sample-based (site-based) CV performance of the nighttime model during daytime periods shows R^2 = 0.90 (0.87) and RMSE = 13.56 (15.11) $\mu g/m^3$. In the sample-based CV, the RMSE of the daytime model is 12.1% lower, while in the site-based CV, it is 8.9% lower. The nighttime model is constructed to address the problem of missing O_3 concentration data at night, and although its performance is

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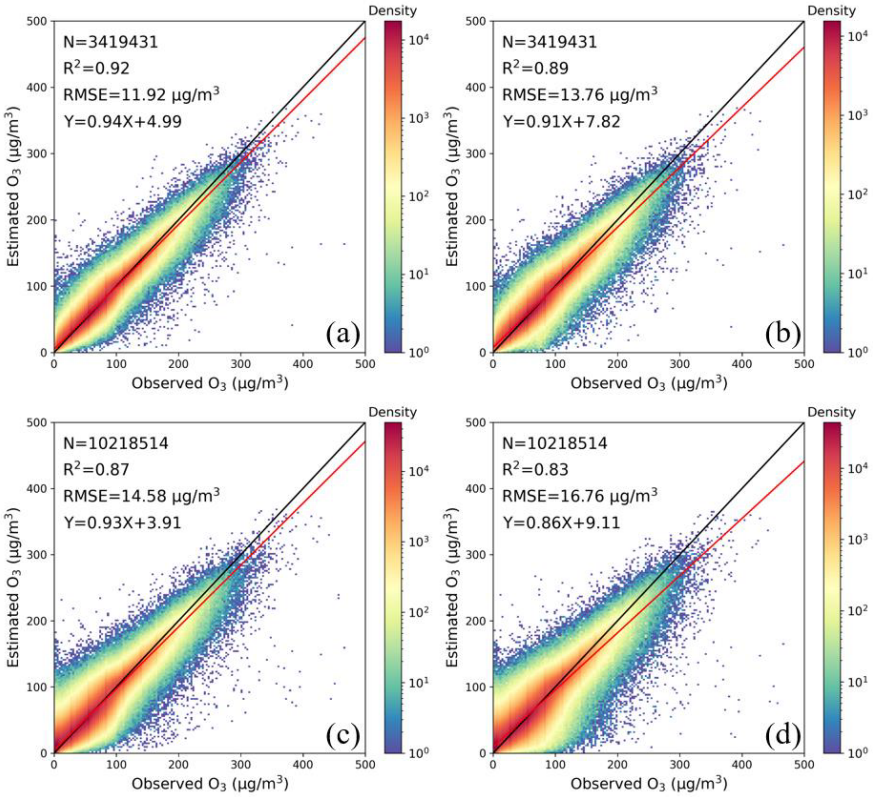


Fig. 3. Ten-fold CV results are (a) and (b) sample-based and site-based CV results of the daytime model, respectively and (c) and (d) sample-based and site-based CV results of the nighttime model, respectively.

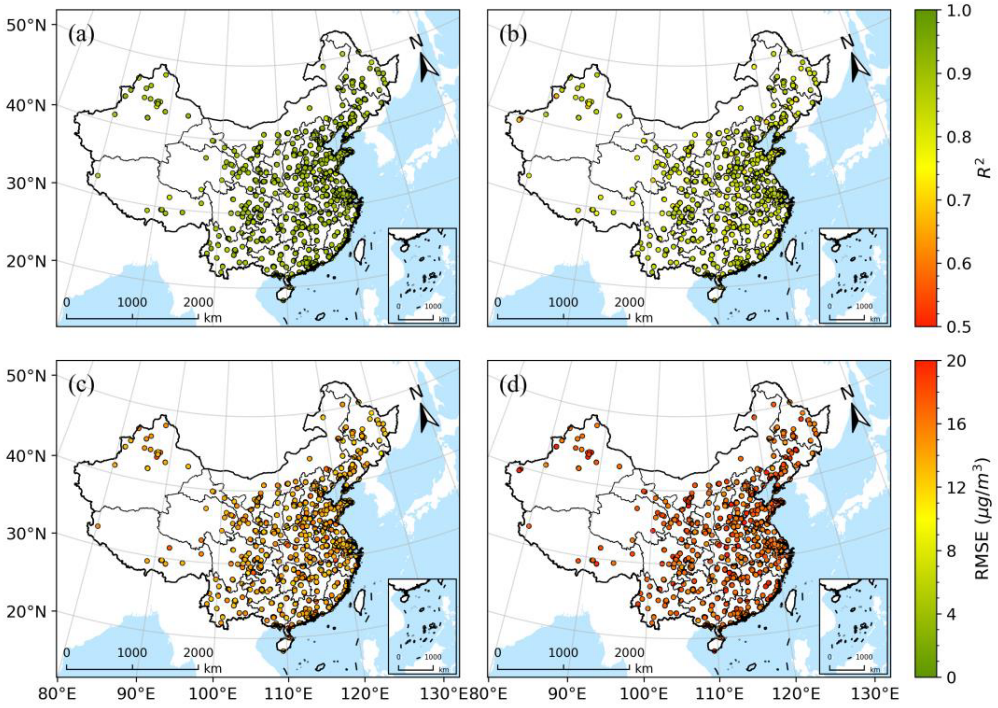


Fig. 4. Performance of daytime and nighttime models at each site are (a) and (b) R^2 values of the daytime model and the nighttime model, respectively, and (c) and (d) RMSE values of the daytime model and the nighttime model, respectively.

relatively lower compared to the daytime model, it still has a high overall performance. All these results demonstrate the effectiveness of the proposed framework in generating 24-h O_3 concentration data.

Fig. 4 shows the R^2 and RMSE values of the daytime and nighttime models at each site. It is found that the daytime model generally performs better than the nighttime model, with higher mean R^2 values (0.89 versus 0.83). Moreover, the

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TABLE II

HIGH-VALUE PERFORMANCE OF THE MODELS FOR THE CASE WITH SAMPLE WEIGHTING, THE VALUES IN PARENTHESES INDICATE THE PERCENTAGE IMPROVEMENT/REDUCTION IN R^2 /RMSE COMPARED TO THE CASE WITHOUT SAMPLE WEIGHTING

Model	High-value criterion	Without sample weighting		With sample weighting	
		R^2	RMSE ($\mu g/m^3$)	R^2	RMSE ($\mu g/m^3$)
Daytime model	$>100\mu g/m^3$	0.78	14.65	0.83 (+6.4%)	12.61 (-13.9%)
	$>120\mu g/m^3$	0.72	16.32	0.77 (+6.9%)	14.04 (-14.0%)
	$>160\mu g/m^3$	0.55	21.54	0.61 (+10.9%)	18.55 (-13.9%)
Nighttime model	$>100\mu g/m^3$	0.68	19.20	0.77 (+13.2%)	14.52 (-24.4%)
	$>120\mu g/m^3$	0.61	21.14	0.72 (+18.0%)	15.85 (-25.0%)
	$>160\mu g/m^3$	0.44	25.93	0.57 (+29.5%)	20.10 (-22.5%)

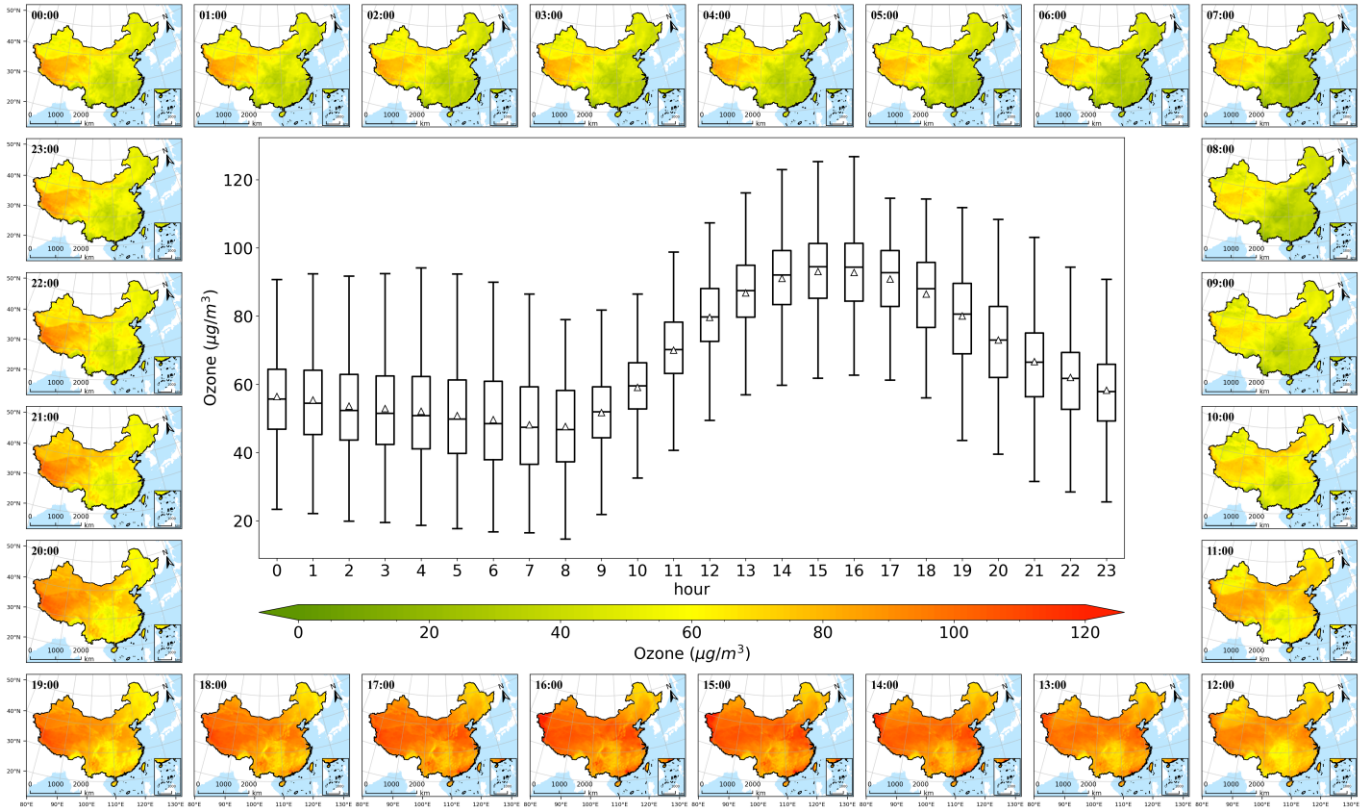


Fig. 5. Hourly average mapping of O_3 across China in 2021. The triangle symbol in the box plot represents the mean value.

daytime model has a relatively consistent high performance among different sites across China, and even in western China, where the sites are sparse, the performance of the daytime model remains high. The nighttime model has high performance in eastern China, where the sites are densely populated, but there is still room for improvement in other areas where the sites are sparse. Overall, the spatial distribution of R^2 and RMSE indicates that both the daytime and nighttime models perform well and are suitable for nationwide estimation of O_3.

2) High-Value Estimation Performance: To evaluate the effectiveness of the sample-weighted approach in mitigating high-value underestimation, different high-value criteria are set (including 100, 120, and 160 $\mu g/m^3$), and the model performance in the high-value samples in the sample-based CV is compared. The results are shown in Table II. Under different high-value criteria, the estimation performance for high-value samples is improved. For 100 $\mu g/m^3$ as the high-value criterion, the high-value samples' R^2 of the daytime model improves by 6.4%, and the RMSE decreases by 13.9%, and for 120 $\mu g/m^3$ as the high-value criterion, the R^2 increases by 6.9%, and the RMSE decreases by 14.0%, and for 160 $\mu g/m^3$ as the high-value criterion, R^2 increases by 10.9%, and the RMSE decreases by 13.9%. This improvement is even more significant in the nighttime model, where the proposed sample-weighted approach increases the high-value R^2 by 13.2% and decreases the RMSE by 24.4% for the high-value criterion of 100 $\mu g/m^3$, increases the R^2 by 18.0% and decreases the RMSE by 25.0% for the high-value criterion of 120 $\mu g/m^3$,

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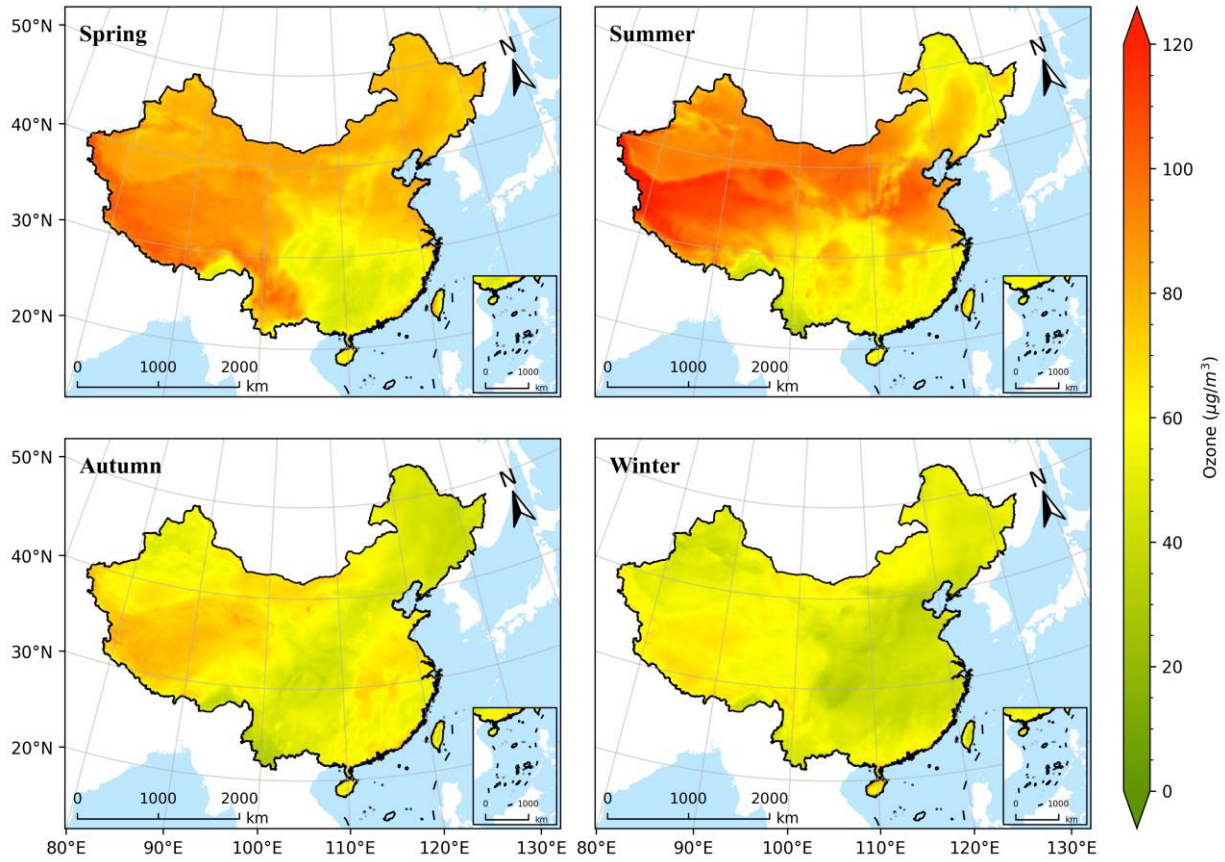


Fig. 6. Seasonal O_3 concentrations across China in 2021. March, April, and May are considered as spring, June, July, and August as summer, September, October, and November as fall, and December, January, and February as winter in this study.

and increases the R^2 by 29.5% and decreases the RMSE by 22.5% for the high-value criterion of 160 $\mu g/m^3$.

B. O_3 Mapping and Spatial-Temporal Analysis

The proposed models are used to estimate hourly surface O_3 concentrations for China. Fig. 5 shows the hourly average mapping results of O_3 over China in 2021. Temporally, the inverted O_3 concentrations reach the lowest value at 08:00 (47.61 $\pm$ 12.55 $\mu g/m^3$), gradually increase between 08:00 and 16:00, and reach the highest value at 16:00 (92.77 $\pm$ 11.32 $\mu g/m^3$), then gradually decrease between 16:00 and 08:00 of the next day. This diurnal pattern may be related to factors such as human activities and changes in solar radiation throughout the day. During the daytime, human activities are more intensive and solar radiation is stronger, leading to increased O_3 production and gradually rising concentrations. At night, O_3 concentrations are generally lower [37], [53], [54], due to the cessation of photochemical production as well as the dissipation of O_3. Spatially, between 08:00 and 13:00, O_3 concentrations in central China rise faster than in other regions, and between 13:00 and 16:00, O_3 concentrations rise to higher levels in the whole of China, with O_3 concentrations in the eastern coast of China being higher than in other regions. Between 16:00 and 08:00 of the next day, O_3 concentrations gradually decrease throughout China, with eastern China decreasing faster than western China.

Fig. 6 shows the seasonal averages of O_3 across China in 2021. China has higher O_3 concentrations during spring and summer, with an average of 76.82 $\pm$ 12.85 $\mu g/m^3$ and 80.86 $\pm$ 18.93 $\mu g/m^3$, respectively. O_3 concentrations are lower in fall and winter, with an average of 57.93 $\pm$ 10.69 $\mu g/m^3$ and 51.92 $\pm$ 8.50 $\mu g/m^3$, respectively. The reason for this seasonal pattern may lie in the fact that O_3 is a photochemical pollutant. High temperatures, strong solar radiation, and a high atmospheric oxidation capacity favor chemical reactions that produce O_3. Overall, spring and summer in China are more conducive to O_3 formation, whereas autumn and winter are relatively less favorable, resulting in lower O_3 concentrations [55]. Spatially, the high O_3 concentrations in spring are mainly reflected in the fact that the O_3 concentrations in most regions of China are above 80 $\mu g/m^3$, and only southern China has lower O_3 concentrations, mostly below 70 $\mu g/m^3$. In summer, the high concentrations in eastern and northern China are mainly above 100 $\mu g/m^3$, while the O_3 concentrations in western China are still above 80 $\mu g/m^3$, and the O_3 concentrations in southern China are lower, around 70 $\mu g/m^3$. Compared with the O_3 concentrations in summer, O_3 concentrations decrease in the fall, with most regions having O_3 concentrations at 60 $\mu g/m^3$, and only some western, northern, and eastern regions have O_3 concentrations above 60 $\mu g/m^3$. O_3 concentrations in these regions also decrease to around 60 $\mu g/m^3$ and below in the winter. The Tibetan

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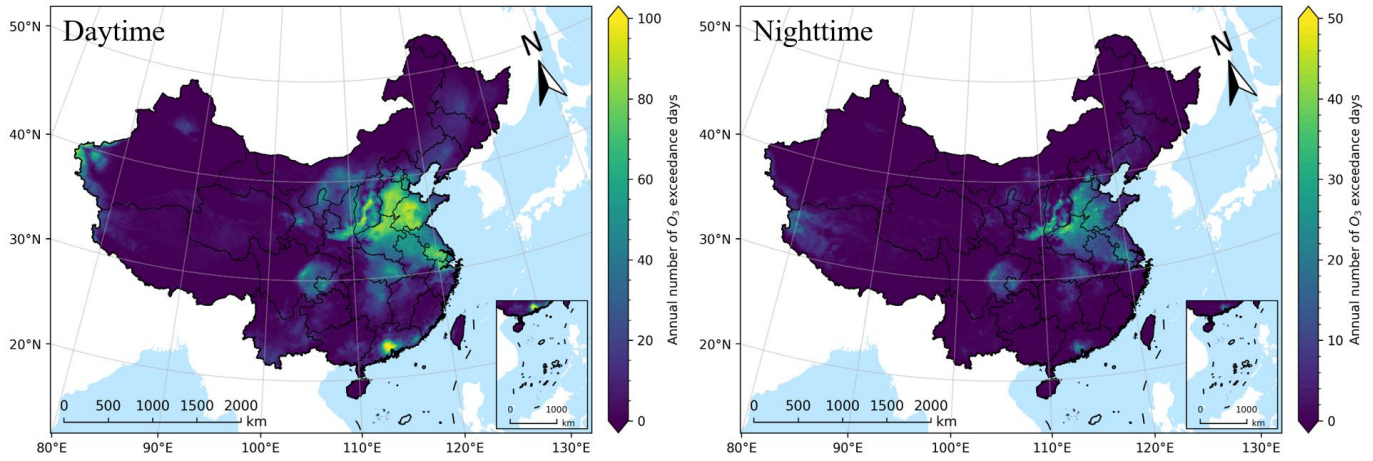


Fig. 7. Distribution of the number of days with excessive O_3 over China in 2021. The number of exceedance days during the daytime is also included for comparison. Daytime statistics use the same rule: one hour exceeding 160 $\mu g/m^3$ makes the day an exceedance.

Plateau exhibits high O_3 concentrations, likely due to the more frequent transport of stratospheric O_3 associated with its high elevation [56].

C. Discussion

1) Importance of Nighttime O_3 Estimation: The nighttime O_3 concentration data cannot be inverted in some previous studies [18], [20], [28], [29]. This limitation can lead to higher O_3 concentrations when calculating daily averages. Specifically, because O_3 concentrations are generally higher during the day than at night, averaging using only daytime data can result in high averages. In our study, it is found that the mean value calculated using only all the daytime (09:00–16:00 local time) inversions is 77.99 $\mu g/m^3$, while the mean value calculated using all the full-day inversions is 66.98$\mu g/m^3$, and the lack of nighttime O_3 concentration data leads to a 16.4% bias in the statistical results.

In addition, we count the number of days with any hour of nighttime O_3 concentrations exceeding 160 $\mu g/m^3$ from 19:00 to 06:00 of the next day in 2021 across China (as shown in Fig. 7). The nighttime O_3 concentration exceedance is more serious in Henan, Hebei, and Shandong, and also exists in a certain range of areas such as Beijing, Tianjin, Shanxi, Jiangxi, and Jiangsu. In addition, the nighttime O_3 exceedances are found in Chengdu, the Greater Bay Area of Guangdong, Wuhan, Changsha, and the western region of Tibet. The results in Fig. 7 may appear inconsistent with the high values over the Tibetan Plateau shown in Figs. 5 and 6; however, this difference arises because Fig. 7 considers exceedances of > 160 $\mu g/m^3$, which are relatively rare over the Tibetan Plateau, whereas exceedances of > 120 $\mu g/m^3$ occur much more frequently. In summary, the lack of nighttime O_3 data will lead to the lack of monitoring of O_3 pollution at night, and the inversion of the 24-h O_3 concentration data is important to improve the performance of the daily average and the monitoring of O_3 concentrations.

We track a severe O_3 pollution outbreak event experienced in eastern China between 0:00 and 05:00 Beijing time on August 10, 2021 (as shown in Fig. 8). According to the inversion data, the O_3 concentrations are at a high level in Shandong, Shanxi, Henan, Hebei, Tianjin, Jiangsu, and Anhui at 0:00 h on that day. The O_3 concentrations increase rapidly in the following hours and reach their peak at 04:00 and 05:00, when Shandong, Tianjin, Hebei, Henan, and Jiangsu are already covered by high O_3 concentrations. The northern region of Anhui is also seriously affected, while the nearby Shaanxi and Liaoning are also polluted. Although the ground site data also observe this O_3 pollution outbreak to some extent, the continuous inversion results have a better performance in showing the dynamics of polluted areas. During this O_3 episode, O_3 shows the strongest correlation with NO_2 (0.43) and a moderate negative correlation with BLH (−0.26), while its correlations with wind speed, temperature, and other meteorological variables are negligible. These results suggest that the episode is likely influenced by the lowering of the BLH, which facilitates the downward mixing of NO_2 and O_3 [57]. More detailed investigations will further require the application of atmospheric chemistry transport models.

2) Temporal Generalization Ability of the Models: To further assess the temporal generalization of the models, we introduce an additional temporal-based validation. Specifically, we train the models using data from 2021 and test them with data from January to June 2022, and the results are presented in the Supplementary Fig. S3. The findings demonstrate that our models exhibit strong predictive capability. In particular, the daytime model achieves an R^2 of 0.71 and an RMSE of 23.52 $\mu g/m^3$, while the nighttime model achieves an R^2 of 0.62 and an RMSE of 25.76 $\mu g/m^3$. Furthermore, the daytime model outperforms the nighttime model during the daytime periods (the nighttime model yields an R^2 of 0.65 and an RMSE of 26.4 $\mu g/m^3$ during the daytime periods). Moreover, we summarize the monthly performance metrics in Supplementary Table S3. The results indicate that the models maintain relatively stable performance over time, highlighting the robustness of our modeling approach.

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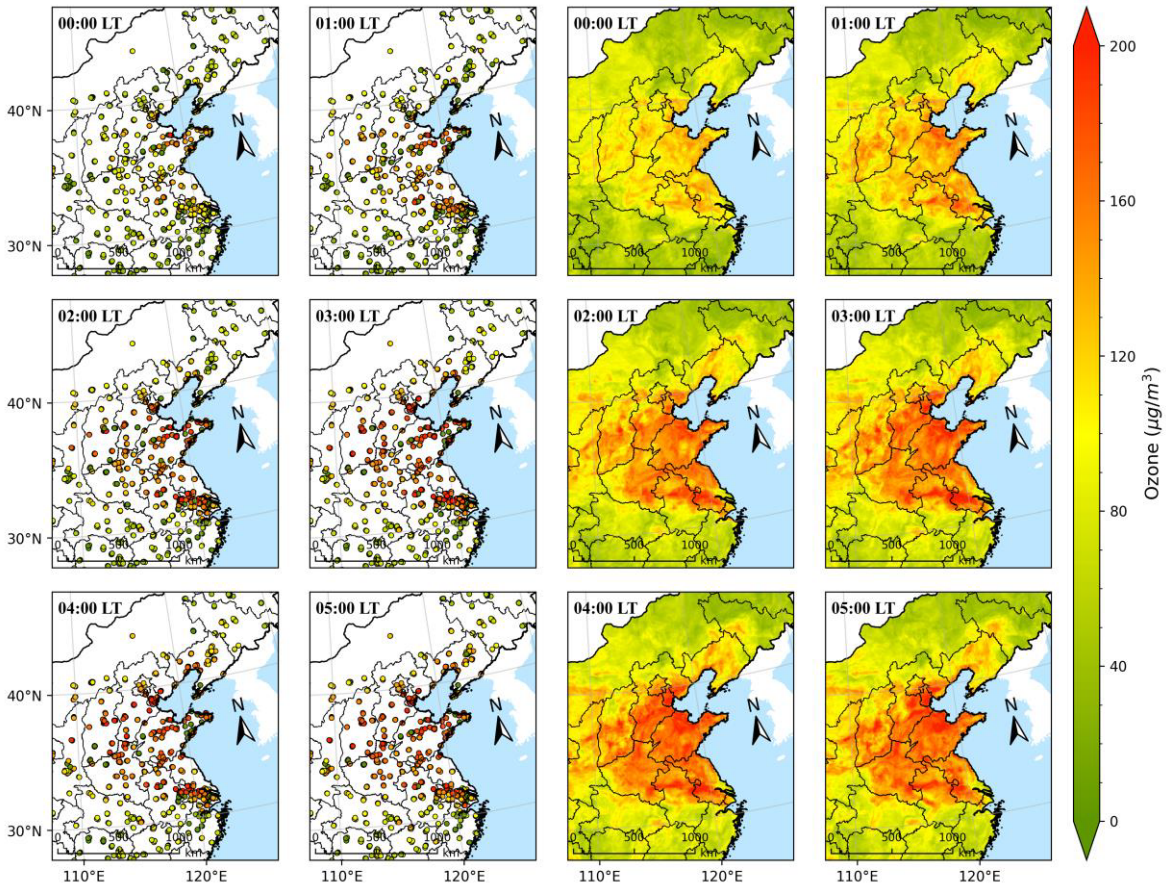


Fig. 8. O_3 concentrations mapping during a pollution period on August 10, 2021.

3) Comparison With Other Studies: We further compare this study with previous works on hourly O_3 estimation, as summarized in Table III. In terms of temporal resolution, hourly O_3 estimation studies provide finer monitoring than daily estimations; however, studies achieving continuous 24-h coverage remain relatively scarce. Methodologically, this study proposes a sample-weighted machine-learning approach, which effectively alleviates the underestimation of high O_3 concentrations. Regarding model performance, comparable performance (R^2 values generally ranging from 0.8 to 0.9) has been reported, all demonstrating strong capability in estimating surface O_3 concentrations. However, substantial differences exist in terms of study periods, data preprocessing, and validation strategies, which makes direct comparisons difficult. Therefore, the performance metrics presented here should be regarded as a reference only. In summary, the sample-weighted LightGBM model proposed in this study generates seamless 24-h surface O_3 datasets across China and serves as an important complement to existing research.

4) Limitations: Although our study proposes a 24-h O_3 estimation framework based on satellite observations and sample-weighted machine learning, and successfully generates seamless 24-h O_3 concentration data across China, several limitations remain. First, the estimation is limited to the year 2021. Extending this framework to produce long-term, seamless 24-h O_3 data warrants further investigation. Second, real-time 24-h monitoring of O_3 concentrations is crucial. However, the current study still faces several challenges in achieving real-time monitoring. For example, the ERA5 reanalysis data can only be obtained with a time delay. Although machine-learning methods offer high prediction efficiency, real-time monitoring remains difficult due to the limitations of the input data. To address this limitation, future work could incorporate near-real-time datasets as model inputs to enable real-time estimation of 24-h O_3 concentrations. Third, the proposed methodological framework is relatively complex, requiring the construction of separate models for daytime and nighttime periods, followed by the fusion of the two outputs. Since the nighttime model is capable of estimating 24-h O_3 concentrations, it raises the question of whether a single model based on nighttime features could suffice. However, it should be noted that the daytime model achieves better performance than the nighttime model, which justifies the need for separate modeling during daytime hours. Finally, the use of two independent models for different periods may raise concerns regarding temporal consistency. According to the results (Supplementary Figs. S4–S6), some temporal inconsistencies are indeed observed; however, they remain within an acceptable range. This may be attributed to the fact that the nighttime model is trained using 24-h station

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TABLE III
COMPARISON BETWEEN DIFFERENT STUDIES

Study	Method	Study period	Study area	Temporal resolution	Model performance
Cheng et al. [37]	4-Dimensional Space-Time Extra-Trees	2019	China	24-hourly	Sample CV R^2: 0.89, RMSE: 16.35 $\mu g/m^3$
Yan et al. [28]	Deep neural network	2017-2019	China	Hourly (09:00-17:00 LT)	Sample CV R^2: 0.86 RMSE: 19.13 $\mu g/m^3$
Chen et al. [19]	Deep forest	2015.12-2021.08	China	Hourly (10:00-15:00 LT)	Sample CV R^2: 0.86–0.92, RMSE: 12.97–14.05 $\mu g/m^3$
Kim et al. [29]	cLearGBM and cLoudGBM	2021-2022	East Asia	24-hourly	Sample CV R^2: 0.90 (0.87) RMSE: 8.77 (9.44) ppm
Zhang et al. [20]	Bagged-tree	2018.03-2019.02	China	Hourly (09:00-16:00 LT)	Sample CV RMSE: 15.8 $\mu g/m^3$
Wang et al. [18]	SGLboost	2017-2019	China	Hourly (10:00-15:00 LT)	Space CV R^2: 0.85 RMSE: 19.04 $\mu g/m^3$
Gao et al. [58]	LightGBM	2022	China	Hourly (08:00-19:00 LT)	Sample CV R^2: 0.96, RMSE: 9.22 $\mu g/m^3$
This study	Sample-weighted LightGBM	2021	China	24-hourly	Daytime (Nighttime) sample CV R^2: 0.92 (0.87), RMSE: 11.92 (14.58) $\mu g/m^3$

Note. cLearGBM: Clear LightGBM, cLoudGBM: cloudy-sky LightGBM, SGLboost: Self-adaptive Geospatially Local scheme based on Categorical boosting.

data, such that daytime observations also contribute to the nighttime model training, thereby facilitating a reasonable connection between the two models. In this study, we adopt two separate models to achieve higher accuracy rather than relying on a single nighttime model, which would provide better temporal consistency. This highlights an important direction for future research—how to improve both model accuracy and temporal consistency simultaneously. Despite this limitation, our approach attains higher accuracy during daytime, and the results within the daytime and nighttime periods are internally consistent. Moreover, few studies have estimated 24-h O_3 concentrations, and thus the proposed framework still offers valuable contributions.

V. CONCLUSION

In this study, the daytime and nighttime models based on LightGBM are constructed using Himawari-8 satellite data, surface temperature data, ERA5 reanalysis data, and EAC4 reanalysis data to successfully invert the seamless 24-h surface O_3 concentration data for the entire year of 2021 in China. The main contributions are as follows:

1) Seamless 24-h surface O_3 concentration data for China in 2021 are constructed. Given the absence of nighttime O_3 data and the inability of O_3 data to cover the whole of China, we construct a daytime model based on SWR data and a nighttime model based on surface temperature data, and integrate the inversion results of the two models to obtain seamless 24-h O_3 data.
2) The sample-weighted approach is adopted to effectively alleviate the high-value underestimation results in the O_3 inversion process. By setting the larger weights for high-value samples, the proposed approach results in a 6.4% R^2 boost and 13.9% RMSE decrease for the high-value samples of the daytime model, and a 13.2% R^2 boost and 24.4% RMSE decrease for the high-value samples of the nighttime model.

This study generates seamless 24-h O_3 concentration data across China by integrating multisource data and constructing a fusion framework combining daytime and nighttime models, offering significant application potential for pollution prevention and control.

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